

RAMCO INSTITUTE OF TECHNOLOGY
Department of Computer Science and Engineering
Academic Year: 2019 - 2020 (Even Semester)

Degree, Semester & Branch: VI Semester B.E. CSE

Course Code & Title: CS8602 Compiler Design

Practice's followed: Learning by Teaching

Type of Activity: Active Learning

Topic: Minimized DFA

Date: 10-01-2020

Objectives:

O1: To make the students to learn various phases of a compiler and obtain the knowledge about lexical analyzer.

Justification for choosing the particular topic:

Minimized DFA is an important problematic concept. This topic was repeatedly asked in university examinations. I have chosen this topic to make the students must well know this concept.

Activity Description:

Learning by Teaching is an educational approach to problem-based learning.

Steps:

1. The topic Minimized DFA was given to Ms.RajaRajeswari of Third Year CSE one week before.
 2. She learned the given topic using the tabulation method from the material.
 3. She teaches the lessons to other students in the class.
 4. First she took a regular expression and convert it into DFA.
 5. She took the same DFA Diagram, and ask the students to draw a table for all pair (final state, non final state) of states and make a mark on a table.
 6. The students find out the unmarked pair, that pair gives the minimal number of state.
 7. The Transition table modified based on that Minimized DFA was drawn by the students.
- During her lecture she solved one example problem clearly.

A student solves the example problem with all the implementation steps. Finally the answer was discussed in the class.





Reflection critique

Post Implementation:

- This activity helped the student in better understanding and knowledge retention than students who spent the same time re-studying.
- It improves the student's efficacy, confidence, communication skills, and taking chances in the public.

Observation Made:

- All the students have performed well they solved the problem of minimized DFA.
- One or two students difficult to understand some steps.

Steps taken for challenges:

- Additional problems were discussed.
- For practicing the method additional problems were given as a home work.

Relevance of Program Outcomes

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
O1	2	2	2					1	2	2		3

References:

https://en.wikipedia.org/wiki/Learning_by_teaching

<https://digest.bps.org.uk/2018/05/04/learning-by-teaching-others-is-extremely-effective-a-new-study-tested-a-key-reason-why/>

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